



balanced equations

Name:

Date:

Which of the following is correctly balanced?

- A. $\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2$
- B. $2 \text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2$
- C. $2 \text{Fe} + 3 \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2$
- D. $\text{Fe} + 3 \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2$

Correct Answer: A B C D

Explanation:

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A student has a 30 cm³ sample of dilute hydrochloric acid.
The concentration of the acid is 1 mol dm⁻³.
He wants the acid to have a concentration of 0.1 mol dm⁻³.
How much water does he need to add to the 30 cm³ sample of acid?

A 30 cm³ **B** 270 cm³ **C** 300 cm³ **D** 330 cm³

Correct Answer: A B C D

Explanation:

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Which symbol equation for the reaction between sodium carbonate (Na₂CO₃) and hydrochloric acid (HCl) correctly represents the word equation below?
Sodium carbonate (solid) + Hydrochloric acid (solution) → Sodium chloride (solution) + Carbon dioxide (gas) + Water (liquid)

A $\text{Na}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + 2\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$

B $\text{Na}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$

C $\text{Na}_2\text{CO}_3(\text{s}) + \text{HCl}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$

D $2\text{Na}_2\text{CO}_3(\text{s}) + 4\text{HCl}(\text{aq}) \rightarrow 4\text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$

Correct Answer: A B C D

Explanation:

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Oxygen forms O^{2-} ions, lithium forms Li^{+} ions.
What is the correct chemical formula for the ionic compound lithium oxide?

A LiO B LiO_2 C Li_2O_2 D Li_2O

Correct Answer: A B C D

Explanation:

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Which is the correctly balanced equation?

a) $H_2 + Cl_2 \rightarrow HCl$
 b) $C_2H_4 + O_2 \rightarrow 2CO_2 + 2H_2O$
 c) $Ca + H_2SO_4 \rightarrow CaSO_4 + H_2$
 d) $H_2 + 2NO \rightarrow H_2O + N_2$

Correct Answer: A B C D

Explanation:

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The chemical formula for a substance is:
 $Al_2(SO_4)_3$

Which of the following correctly describes the ratio of numbers of atoms of the elements in the formula?

A 2 aluminium atoms : 1 sulfur atom : 4 oxygen atoms B 2 aluminium atoms : 3 sulfur atoms : 4 oxygen atoms
 C 2 aluminium atoms : 3 sulfur atoms : 12 oxygen atoms D 2 aluminium atoms : 3 sulfur atoms : 4 oxygen atoms

Correct Answer: A B C D

Explanation:

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Which is the correctly balanced equation?

a) $2Fe_2O_3 + CO \rightarrow 4Fe + CO_2$
 b) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
 c) $Fe_2O_3 + 3CO \rightarrow 3Fe + 3CO_2$
 d) $Fe_2O_3 + 3CO \rightarrow Fe + 3CO_2$

Correct Answer: A B C D

Explanation:

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Which is the correctly balanced equation?

- a) $2\text{CH}_4 + \text{O}_2 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}$
- b) $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- c) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- d) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

Correct Answer: A B C D

Explanation:

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Which is the correctly balanced equation?

- a) $2\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + 2\text{CO}_2$
- b) $2\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow 2\text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$
- c) $\text{CaCO}_3 + 2\text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{H}_2\text{O} + \text{CO}_2$
- d) $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$

Correct Answer: A B C D

Explanation:

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Which is the correctly balanced equation?

- a) $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$
- b) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
- c) $2\text{Mg} + 2\text{O}_2 \rightarrow 2\text{MgO}$
- d) $3\text{Mg} + 2\text{O}_2 \rightarrow 4\text{Mg}_2\text{O}$

Correct Answer: A B C D

Explanation:

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Which is the correctly balanced equation?

- a) $\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- b) $2\text{H}_2 + 2\text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- c) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- d) $3\text{H}_2 + 2\text{O}_2 \rightarrow 4\text{H}_2\text{O}$

Correct Answer: A B C D

Explanation:

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Which is the correctly
balanced equation?

- a) $\text{N}_2 + \text{H}_2 \rightarrow 2\text{NH}_3$
- b) $2\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$
- c) $3\text{N}_2 + 3\text{H}_2 \rightarrow 3\text{NH}_3$
- d) $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$

Correct Answer: A B C D

Explanation:

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